Real Time Air Quality Analysis - INDIA

1. Objectives:

The objective of this project is to analyze air quality data collected from various locations across India using Excel and Power BI. The project aims to identify pollution trends, compare pollutant levels across states, and create an interactive dashboard for better understanding and decision-making.

2. Dataset Description:

The dataset contains air quality information collected from different monitoring stations across India.

It includes:

- *State*
- *City*
- *Station*
- *Pollutant ID*
- *Pollutant Minimum*
- *Pollutant Maximum*
- *Pollutant Average*
- *Latitude*
- *Longitude*
- *Last Update*

3. Tools Used:

- ➢ *Microsoft Excel*
- ➢ *Power BI Desktop*

4. Data Pre-processing in Excel:

The following data cleaning and transformation tasks were performed:

- ✓ *Removed duplicate records*
- ✓ *Replaced "NA" values with blanks*
- ✓ *Checked missing values*
- ✓ *Standardized column formats*
- ✓ *Created Pivot Tables for summary analysis*

5. Data Transformation in Power BI:

The dataset was transformed using Power Query by:

- *Importing cleaned Excel data*
- *Converting appropriate data types*
- *Handling null values*
- *Loading the transformed data into Power BI*

6. Dashboard Features:

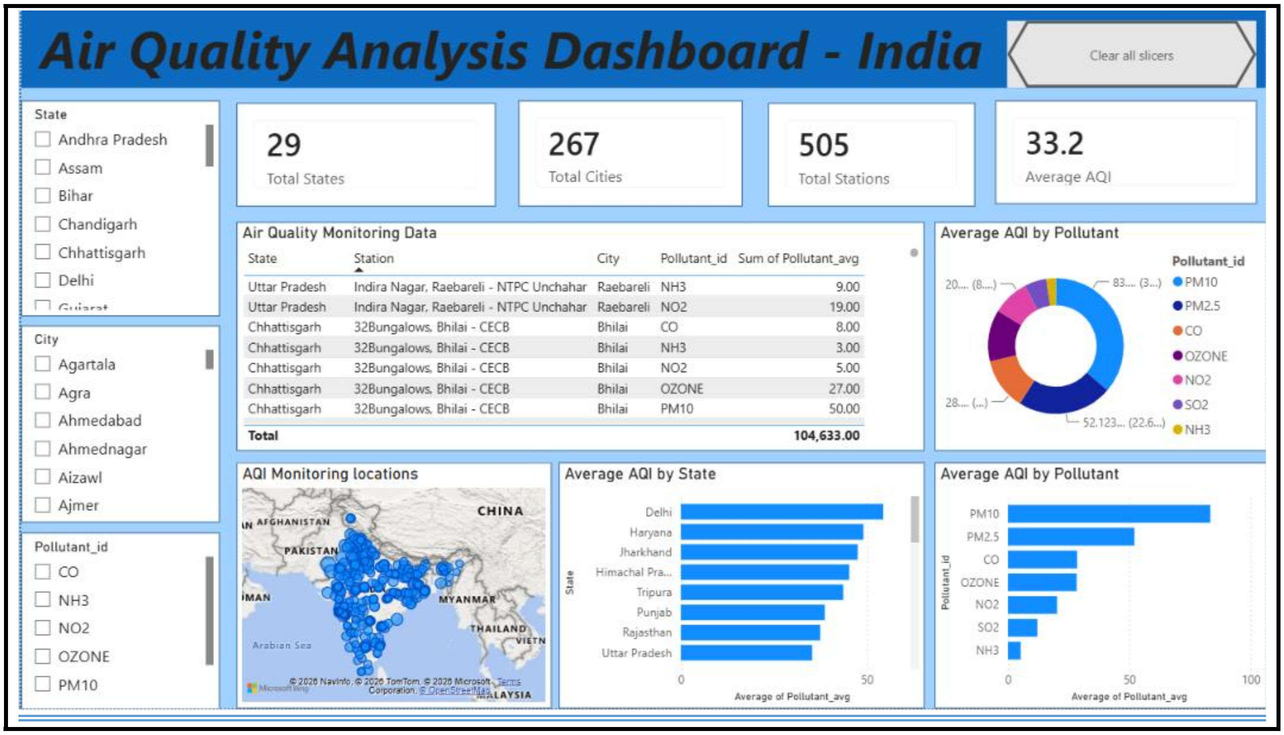
The dashboard includes:

- ***KPI Cards:***
    - *Total States*
    - *Total Cities*
    - *Total Stations*
    - *Average AQI*
- ***Slicers:***
    - *State*
    - *City*
    - *Pollutant_ID*
- ***Charts:***
    - *Average AQI by State*
    - *Average AQI by Pollutant*
    - *Donut Chart of Pollutants*
    - *Monitoring Locations Map*
    - *Air Quality Data Table*

7. Key Insights:

- *The dataset contains 29 States, 267 Cities, and 505 Monitoring Stations.*
- *PM10 has the highest average pollution level.*
- *PM2.5 is the second highest pollutant.*
- *NH3 has the lowest average pollution level.*
- *The dashboard allows users to filter data by State, City, and Pollutant.*
- *The dashboard shows that Delhi has one of the highest average AQI values among the states in the dataset.*

8.Dashboard:



Dashboard Description

- *The Air Quality Analysis Dashboard – India was developed in Power BI to analyze air quality data collected from various monitoring stations across India. The data was cleaned and transformed using Power Query before creating the dashboard.*
- *The dashboard includes:*
- *KPI Cards displaying the total number of States (29), Cities (267), Stations (505), and the Average AQI (33.2).*
- *Slicers for State, City, and Pollutant to enable interactive filtering.*
- *A Table Visual showing State, Station, City, Pollutant, and Average AQI.*
- *A Donut Chart showing the distribution of average AQI by pollutant.*
- *A Map Visual displaying the geographical locations of monitoring stations across India.*
- *A Bar Chart comparing the average AQI across different states.*
- *A Bar Chart comparing the average AQI for different pollutants.*
- *A Clear All Slicers button to quickly reset all applied filters.*

The dashboard provides an interactive and user-friendly way to explore air quality data and compare pollution levels across different states, cities, and pollutants.

9. Conclusion

This project successfully analyzed air quality data from various locations across India using Microsoft Excel and Power BI. Excel was used for data cleaning and pre processing, while Power BI was used to transform the data and create an interactive dashboard.It provides a clear overview of air quality by displaying key metrics such as the total number of states, cities, monitoring stations, and the average AQI. It also helps compare pollution levels across different states and pollutants through interactive charts, maps, and filters.This project demonstrates how data analytics can support better understanding of air quality patterns and assist in making informed environmental decisions.